Lecture 24: Power Series (10.6)

OBJECTIVES:

L24
- Define power series
- Compute interval and radius of convergence for a power series

L25
L26.
- Find a power series representation for a function
- Modify the formula $\boxed{\frac{1}{1-x} = \sum_{0}^{\infty} x^n}$ $-1<x<1$ to find the power series expansions of other functions

L27
L28
- Find Taylor series for $f(x)$ centered at a
- Express integrals using series
- Approximate function values using Taylor Polynomial

L29
- Find n^{th} derivatives $f^{(n)}$

summary of power series.

Power Series

A **power series centers at** 0 is an infinite series of the form

constants/coefficients.

$$\sum_{n=0}^{\infty} \boxed{c_n} x^n = \underbrace{c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n}_{\text{n-th degree polynomial}} + \underbrace{c_{n+1} x^{n+1} + \cdots}_{\text{terms continues}}$$

$(x-0)^n$

When the **power series centers at** a, it has the form

$$\sum_{n=0}^{\infty} c_n (x - \boxed{a})^n = \underbrace{c_0 + c_1(x-a) + \cdots + c_n(x-a)^n}_{\text{nth degree polynomial}}$$

$$+ \underbrace{c_{n+1}(x-a)^{n+1} + \cdots}_{\text{terms continues}}$$

ex. Which ones below are power series?

	y/n
$x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots = \sum_{n=1}^{\infty} \frac{1}{n!} x^n$	Y
$x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots = \sum_{n=1}^{\infty} \frac{1}{n} x^n$	Y
$\frac{1}{x} - x + x^2 + \cdots$	N.
$x + 2!x^2 + 3!x^3 + \cdots = \sum_{n=1}^{\infty} n!\, x^n$	Y
$(x-1) + (x-1)^2 + (x-1)^3 + \cdots$	Y

$= \sum_{n=1}^{\infty} (x-1)^n$

center at 0.

centers at 1.

Some values of x will cause the series to converge, some don't. We call the values of x for which a power series converges its **interval of convergence**.

① ratio test $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$ $\longrightarrow$ convergence.

② root test $\lim_{n\to\infty}\sqrt[n]{|a_n|}<1$

ex. For what values of x would the series converge? $\displaystyle\sum_{n=1}^{\infty}\frac{x^n}{n!}$, $\displaystyle\sum_{n=1}^{\infty}(x-1)^n$, $\displaystyle\sum_{n=1}^{\infty}x^n n!$

ratio.

① $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{\cancel{n+1}}}{\underbrace{\cancel{(n+1)!}}_{n+1}}\cdot\frac{\cancel{n!}}{\cancel{x^n}}\right| = \lim_{n\to\infty}\frac{|x|}{n+1} = 0<1$$

I.O.C $=(-\infty,\infty)$

$R=\infty$

root

② $$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}|x-1| = |x-1| < \boxed{1}$$

$-1<x-1<1$

$0 < x < 2$

check $x=0$: $\sum_{n=1}^{\infty}(0-1)^n$

$= \sum_{n=1}^{\infty}(-1)^n$ div. TFD

$x=2$

$\sum_{n=1}^{\infty}(x-1)^n$: geometric series

$-1<r=x-1<1$

$0<x<2$

I.O.C $=(0,2)$

$R=1$

ratio.

③ $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{\cancel{n+1}}\cdot\overbrace{\cancel{(n+1)!}}^{n+1}}{\cancel{x^n}\cdot\cancel{n!}}\right| = \lim_{n\to\infty}(n+1)\cdot|x|$$

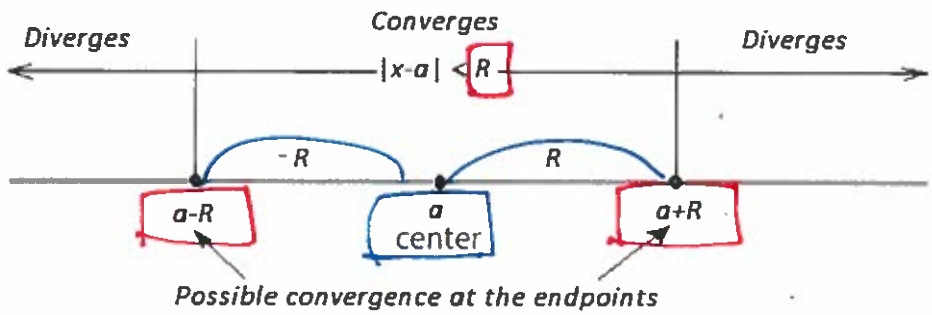
$$= \begin{cases}\infty & \text{if } x\neq 0\\ 0<1 & \text{if } x=0\end{cases}$$

$R=0$

I.O.C. $=\{0\}$

Radius of Convergence

Every power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has a radius of convergence R.



When a power series converges over a finite interval, the constant R is half the length of the IOC. The series converges for $|x-a| < R$ and diverges for $|x-a| > R$;

When a power series converges everywhere, we say $R = \infty$;

When a power series diverges, we say $R = 0$, because it diverges everywhere except at the center itself.

Note that the **interval of convergence** is symmetric with respect to the center a and that convergence can happen at end point(s), they must be checked separately.

With a Power series, we like to know for what values of x does it converge.

There are two steps involved in determining the **interval of convergence** of a power series:

Step 1: Find the radius of convergence R (using the Ratio Test, or Root Test in most cases),

Step 2: Check convergence at the end points, if $R \neq 0$ or ∞. (using a test other than Ratio or Root Test to determine the **endpoints** for convergence.)

$\lim_{n\to\infty} \sqrt[n]{n} \to 1$ (L'H)

ex. Find the radius and the interval of convergence for each power series.

$$1. \sum_{n=1}^{\infty} \frac{(2x+1)^n}{n} = (2x+1) + \frac{(2x+1)^2}{2} + \cdots$$

root

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{|2x+1|}{\sqrt[n]{n}} = |2x+1| < 1$$

$$|x+\tfrac{1}{2}| < \boxed{\tfrac{1}{2}} = R$$

Center $a = -\frac{1}{2}$

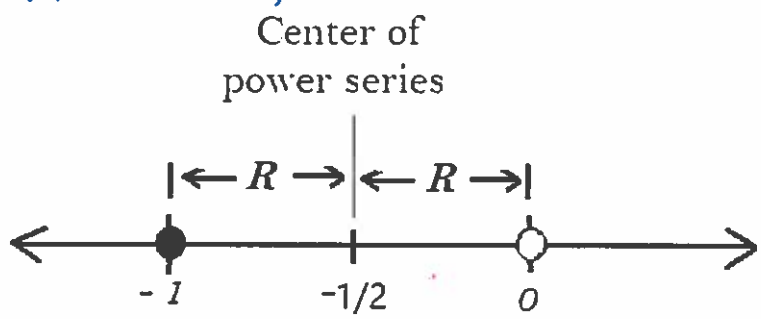
$$-\tfrac{1}{2} < x + \tfrac{1}{2} < \tfrac{1}{2}$$

$$-1 < x < 0$$

Check:

$x=-1$: $\sum_{n=1}^{\infty} \frac{(2(-1)+1)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ conv. by AST.

$x=0$: $\sum_{n=1}^{\infty} \frac{(2\cdot 0+1)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ div. $(p=1)$



Interval of convergence $[-1, 0)$
Radius of convergence $R = 1/2$

$$2.\ \sum_{n=1}^{\infty} \underbrace{\frac{2\cdot 4\cdot 6\cdots 2n}{1\cdot 3\cdot 5\cdots(2n-1)}x^{n}}_{a_n}$$

$n+1$ $\quad$ $n+1$ $\quad$ $n+1$

ratio:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\cancel{2\cdot 4\cdot 6\cdots 2n}\,\boxed{(2n+2)}\,x^{n+1}}{\cancel{1\cdot 3\cdot 5\cdots(2n-1)}\,\boxed{(2n+1)}}\cdot\frac{\cancel{1\cdot 3\cdot 5\cdots(2n-1)}}{\cancel{2\cdot 4\cdot 6\cdots(2n)\cdot x^{n}}}\right|$$

$$= \lim_{n\to\infty}\frac{2n+2}{2n+1}\cdot|x| = |x| < \boxed{1} = R$$

$$-1 < x < 1$$

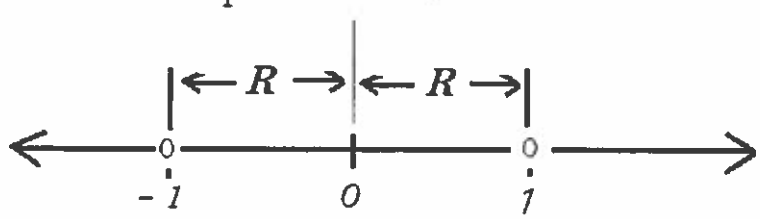
Check:

$x=-1$: $\displaystyle\sum_{n=1}^{\infty}\frac{2\cdot 4\cdot 6\cdots 2n}{1\cdot 3\cdot 5\cdots(2n-1)}\cdot(-1)^n$ div. by TFD.

$$\underset{1}{\underset{\vee}{\left(\frac{2}{1}\right)}}\underset{1}{\underset{\vee}{\left(\frac{4}{3}\right)}}\underset{1}{\underset{\vee}{\left(\frac{6}{5}\right)}}\cdots\underset{1}{\underset{\vee}{\left(\frac{2n}{2n-1}\right)}} > 1 \not\to 0$$

$x=1$: $\displaystyle\sum_{n=1}^{\infty}\frac{2\cdot 4\cdot 6\cdots 2n}{1\cdot 3\cdot 5\cdots(2n-1)}\cdot 1$ div. by TFD.

Center of
power series



Interval of convergence (-1, 1)
Radius of convergence R=1

L24 - 8

NYTI: Find the radius and interval of convergence

root 1. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x+2)^n}{n2^n}$

ratio 2. $\displaystyle\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

root 3. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n}$

ratio 4. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{n \ln n}$

root 5. $\displaystyle\sum_{n=5}^{\infty} \frac{(x-4)^n}{n}$

root 6. $\displaystyle\sum_{n=2}^{\infty} \frac{(3x-2)^n}{n3^n}$

Find the Radius of convergence.

root

7. $\displaystyle\sum_{n=2}^{\infty} \left(\frac{n}{n+1}\right)^{n^2} x^n$

ratio

8. $\displaystyle\sum_{n=2}^{\infty} \frac{n!}{3 \cdot 6 \cdot 9 \cdots (3n)} x^n$

Lecture 24 – NYTI

Find the radius and interval of convergence

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x+2)^n}{n2^n}$ $\quad R=2\ ;\ IOC = (-4, 0]$

root test: $\displaystyle\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{|x+2|}{\sqrt[n]{n}\cdot 2} = \frac{|x+2|}{2} < 1$

$|x+2| < 2\ ,\ R=2$

$-2 < x+2 < 2$

$-4 < x < 0$

$x=-4$: $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-2)^n}{n\cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-1)^n 2^n}{n\cdot 2^n}$

$\displaystyle= \sum_{n=1}^{\infty} \frac{(-1)^{2n-1}}{n} = \sum_{n=1}^{\infty} \frac{-1}{n}$ div. $(p=1)$

$x=0$: $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\cdot 2^n}{n\cdot 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ conv by AST

2. $\displaystyle\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ $\quad R=\infty\ ;\ IOC = (-\infty, \infty)$

ratio test: $\displaystyle\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{x^{2n+2}}{(2n+2)!}\cdot\frac{(2n)!}{x^{2n}}\right| = \lim_{n\to\infty} \frac{x^2}{(2n+1)(2n+2)} = 0 < 1$

for all x

3. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n}$ $\quad R=\infty$; $IOC = (-\infty, \infty)$

root test: $\displaystyle\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} = \frac{|x|}{\ln n} = 0 < 1$ for all x

4. $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{n \ln n}$ $\quad R=1$; $IOC = [-1, 1)$

ratio test: $\displaystyle\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{x^{n+1}}{(n+1)\cdot \ln(n+1)} \cdot \frac{n \ln n}{x^n}\right|$

$$= \lim_{n\to\infty} |x| \cdot \underbrace{\frac{n}{n+1}}_{\downarrow\, 1} \cdot \underbrace{\frac{\ln n}{\ln(n+1)}}_{\downarrow\, L'H\ 1}$$

$= |x| < 1$, $R = 1$

$x=-1$: $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$: conv by AST.

$x=1$: $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$: div. by integral test.

5. $\displaystyle\sum_{n=5}^{\infty} \frac{(x-4)^n}{n}$ $\quad R=1$; $IOC = [3,5)$

root test: $\displaystyle\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{|x-4|}{\sqrt[n]{n}} = |x-4| < 1$, $R=1$

$-1 < x-4 < 1$, $3 < x < 5$

$x=3$: $\displaystyle\sum_{n=5}^{\infty} \frac{(3-4)^n}{n} = \sum_{n=5}^{\infty} \frac{(-1)^n}{n}$ conv by AST.

$x=5$: $\displaystyle\sum_{n=5}^{\infty} \frac{(5-4)^n}{n} = \sum_{n=5}^{\infty} \frac{1}{n}$ div. $(p=1)$

6. $\displaystyle\sum_{n=2}^{\infty} \frac{(3x-2)^n}{n3^n}$ $\quad R=1$; $IOC = [-\frac{1}{3}, \frac{5}{3})$

root test: $\displaystyle\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{|3x-2|}{\sqrt[n]{n}\cdot 3} = \frac{|3x-2|}{3} = \left|x-\frac{2}{3}\right| < 1$

$R=1$

$-1 < x-\frac{2}{3} < 1$

$-\frac{1}{3} < x < \frac{5}{3}$

$x=-\frac{1}{3}$: $\displaystyle\sum_{n=2}^{\infty} \frac{(-1-2)^n}{n\cdot 3^n} = \sum_{n=2}^{\infty} \frac{(-3)^n}{n\cdot 3^n} = \sum_{n=2}^{\infty} \frac{(-1)^n\cdot 3^n}{n\cdot 3^n}$

$\displaystyle= \sum_{n=2}^{\infty} \frac{(-1)^n}{n}$ conv by AST.

$x=\frac{5}{3}$: $\displaystyle\sum_{n=2}^{\infty} \frac{(5-2)^3}{n\cdot 3^n} = \sum_{n=2}^{\infty} \frac{3^n}{n\cdot 3^n} = \sum_{n=2}^{\infty} \frac{1}{n}$ div. $(p=1)$

Find the Radius of convergence.

7. $\displaystyle\sum_{n=2}^{\infty}\left(\frac{n}{n+1}\right)^{n^2} x^n$ $\boxed{R=e}$

root test: $\displaystyle\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\left[\left(\frac{n}{n+1}\right)^{n^2}\right]^{1/n}\cdot|x| = \lim_{n\to\infty}\left(\frac{n}{n+1}\right)^{n}|x|$

$$= \frac{|x|}{e} < 1, \quad |x| < e$$

$$\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^{n} \overset{(1^\infty)}{=} \lim_{n\to\infty} e^{\ln\left(\frac{n}{n+1}\right)^{n}} = \lim_{n\to\infty} e^{\frac{\ln\left(\frac{n}{n+1}\right)}{\frac{1}{n}}} = \lim_{n\to\infty} e^{\frac{\ln n - \ln(n+1)}{\frac{1}{n}}}$$

$$\overset{L'H}{=} e^{\lim_{n\to\infty}\frac{\frac{1}{n}-\frac{1}{n+1}}{-\frac{1}{n^2}}} = e^{\lim_{n\to\infty}\frac{\frac{1}{n^2+n}}{-\frac{1}{n^2}}} = e^{\lim_{n\to\infty} -\frac{n^2}{n^2+n}} = e^{-1}$$

$$= \frac{1}{e}$$

8. $\displaystyle\sum_{n=2}^{\infty}\frac{n!}{3\cdot 6\cdot 9\cdots(3n)}x^n$ $\boxed{R=3}$

ratio test: $\displaystyle\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\overset{n+1}{\cancel{(n+1)!}}\, x^{\cancel{n+1}}}{\cancel{3\cdot 6\cdot 9\cdots(3n)}(3n+3)}\cdot\frac{\cancel{3\cdot 6\cdot 9\cdots(3n)}}{\cancel{n!}\,\cancel{x^n}}\right|$

$$= \lim_{n\to\infty}\frac{(n+1)|x|}{3n+3}$$

$$= \frac{|x|}{3} < 1$$

$$|x| < 3$$